

In particular $F_{\min}(\alpha)$ determines $\mathcal{F}[\alpha]$, and furthermore $\mathcal{F}[\alpha]$ determines $F_{\min}(\alpha)$ up to conjugacy *except* in the case that $\mathcal{F}[\alpha] = \mathcal{A}$: in that case, $F_{\min}(\alpha)$ can be either trivial or atomic. Despite this ambiguity we will refer to either the free factor $F_{\min}(\alpha)$ or the free factor system $\mathcal{F}[\alpha]$ as the *filling support* of α ; in context the exact referent should be clear.

We sometimes use the notation $\mathcal{F}[\alpha; T]$ to highlight the dependence on the free splitting T that contains α . Also, we note that $\mathcal{F}[\alpha]$ is determined by the connected subgraph $\beta_{\text{Id}}^{\circ}(\alpha)$, in that it is the unique smallest free factor system that “carries” the subgroup $\text{Stab}(\beta_{\text{Id}}^{\circ}(\alpha))$. For this reason we sometimes denote $\mathcal{F}[\alpha] = \mathcal{F}[\beta^{\circ}(\alpha; T)]$ and $\text{KR}(\alpha) = \text{KR}(\beta^{\circ}(\alpha; T))$.

Remark. In the statement of Definition 2.8, Case (1) holds if and only if $\beta_{\text{Id}}^{\circ}(\alpha) = \alpha$ if and only if for all $g \neq h \in \Gamma$ the translated paths $g \cdot \alpha$ and $h \cdot \alpha$ do not overlap; in this case the protocomponents of $\beta^{\circ}(\alpha)$ are precisely the translates of α . Also, Case (2) holds if and only if there exists a vertex $V \in \beta_{\text{Id}}^{\circ}(\alpha)$ such that $\text{Stab}(\beta_{\text{Id}}^{\circ}(\alpha)) = \text{Stab}(V)$ and $[\text{Stab}(V)] \in \mathcal{A}$; it follows in this case that $V \in \alpha$, and that $\beta_{\text{Id}}^{\circ}(\alpha) = \bigcup_{g \in \text{Stab}(V)} g \cdot \alpha$.

A continued example. In the example we gave preceding the statement of Theorem 2.4, the protocomponents of $\beta_{\text{Id}}^{\circ}(\alpha)$ are the axes in T of the infinite cyclic free factor $\langle a \rangle$ of $\Gamma = \langle a, b \rangle$, and so the filling support is $\mathcal{F}[\alpha; T] = \{[\langle a \rangle]\}$ which is a nonfull free factor system, despite the fact that the path $\alpha \subset T$ does indeed have an interior crossing of a translate of every natural edge orbit of T : the tree T has only one natural edge orbit, and α crosses three different natural edges of that orbit, the middle one of those three being an interior crossing.

A new example. This example shows that the subgroup $\text{Stab}(\beta_{\text{Id}}^{\circ}(\alpha))$ *need not be* a free factor of $\Gamma \text{ rel } \mathcal{A}$, in contrast to the fact that the stabilizer of every component of the Γ -invariant subforest $\beta(\alpha) \subset T$ is a free factor of $\Gamma \text{ rel } \mathcal{A}$. Consider the rank 4 free group $\Gamma = \langle a, b, c, d \rangle$. Let G be a marked graph with two vertices p, q , with the $\langle a, b \rangle$ rose attached to p , the $\langle c, d \rangle$ rose attached to q , and an edge e from p to q ; we identify $\pi_1(G, p) \approx \langle a, b \rangle * \langle c, d \rangle \approx \langle a, b, c, d \rangle$ by using e as a maximal subtree of G . Let $U = \tilde{G}$ be the universal covering with edge labels lifted from G . Choosing a lift $\tilde{p} \in U$ of p determines the deck transformation action $\Gamma \curvearrowright U$. Let $\tilde{\alpha} \subset U$ be the path with initial vertex $\tilde{p}$ that is labelled by the word $e c d \bar{c} \bar{d} \bar{e} a b \bar{a} \bar{b} e$. Note that $\tilde{\alpha}$ is contained in the U -axis of the infinite cyclic subgroup

$$C = \langle c d c^{-1} d^{-1} a b a^{-1} b^{-1} \rangle < \langle a, b, c, d \rangle$$

Also, $\tilde{\alpha}$ consists of one entire fundamental domain for the action of C on its U -axis, followed by the first e edge of the next fundamental domain. Let T be the free splitting obtained from U by collapsing all edges labelled a, b, c, d , so T has a single natural edge orbit, represented by the T -image of any e -edge of U . Let $\alpha \subset T$ be the image of $\tilde{\alpha}$. Note that α crosses the unique natural edge orbit three times, the middle crossing being an interior crossing. Again α is contained in the axis of C in T , and α consists of one entire fundamental domain of that axis followed by the first edge of the next fundamental domain. Note that no two distinct translates of the axis of C have a common edge in T . It follows that the protocomponent $\beta_{\text{Id}}^{\circ}(\alpha)$ equals the axis of C in T , and that $C = \text{Stab}(\beta_{\text{Id}}^{\circ}(\alpha))$. But C has trivial image under abelianization of $\langle a, b, c, d \rangle$ and hence C is *not* a free factor of $\langle a, b, c, d \rangle$. In fact we have the equation $\mathcal{F}[\alpha; T] = \{[\langle a, b, c, d \rangle]\}$, in other words C fills the group $\langle a, b, c, d \rangle$, and hence α fills T by Proposition 2.9.

The fact that C fills the group $\langle a, b, c, d \rangle$ can be proved using a beautiful transversality argument that we learned from a paper of Stallings [Sta00]. Here are some details of this argument. Consider S